How Computers That Learn From Old Storms Are Helping Forecasters

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When the first **hurricane** of this Atlantic season formed, a computer model that learns from history beat the traditional models at guessing where it would go. In early October, **Hurricane** Isaias made landfall near Destin, Florida, as a Category 2 storm. Traditional models first predicted it would head farther west, toward southeastern Louisiana. But Google DeepMind's artificial intelligence model got it right from the start: the storm would hit the western Florida Panhandle. Meteorologist Michael Lowry calls DeepMind the gold standard model, and says it has been by far the best-performing computer model **forecasters** have ever seen. The National **Hurricane** Center first added the DeepMind model to its official forecasts last year, and by the end of the 2025 season it was already one of the best at predicting a storm's path and strength. Traditional physics-based models use live data on temperature, **humidity**, pressure and wind shear, gathered by satellites and aircraft, then use complicated equations to work out a storm's path and strength. An AI model works differently: it looks for patterns by studying thousands of past storms and **atmospheric** data. A **neural** network is a set of computer programs modeled on the human brain and trained to find patterns in huge amounts of data. That can feel like a black box, because **forecasters** do not always know why it is right or wrong. DeepMind's model and one from the European Centre for Medium-Range Weather Forecasts have proved most accurate so far. The **Hurricane** Center still blends several proven models, both AI and traditional, and humans interpret the results before telling the public. Brennan says **forecasters** turn shifting outputs into clear guidance. DeepMind topped other models on Isaias's path, Brennan says, but did not do as well as the blended forecast on the storm's strength.

Word Treasure

hurricane -- A very large, powerful storm with strong winds that forms over the ocean.
forecasters -- People whose job is to predict what the weather will do.
neural -- Relating to a computer system designed to work a little like a human brain.
atmospheric -- Relating to the air around the Earth.
humidity -- The amount of water vapor in the air.

Background you need

A hurricane is a huge rotating storm that forms over warm ocean water and can bring strong winds and heavy rain to land.

Forecasters try to guess where a storm will go and how strong it will be, so people have time to get ready.

Older models use live measurements and physics equations to work out a storm's path, while a learning model looks for patterns in many past storms.

A neural network is a set of computer programs that learns by finding patterns in very large amounts of data, though people do not always know why it picks a certain answer.

Break it down

LEAD: A computer model that learns from history beat traditional models at guessing where the first hurricane of the season would go.

KEY EVENT: In early October, Hurricane Isaias made landfall near Destin, Florida, as a Category 2 storm.

+- WHO: Google DeepMind's AI model, traditional physics-based models, and the National Hurricane Center.

+- WHAT HAPPENED: DeepMind got the path right from the start, while traditional models first pointed farther west.

+- HOW IT WORKS: Traditional models use live data and equations; the AI model studies patterns in thousands of past storms.

EVIDENCE: Lowry calls DeepMind the gold standard; Brennan says it topped others on path but not on strength versus the blended forecast.

OPEN QUESTION: Why is the AI model sometimes right or wrong, when forecasters do not always know its reasons?

Why it matters

Better storm-path guesses give people more time to prepare, and forecasters still check several models and use human judgment before warning the public.

Test what you remember

1. Where did Hurricane Isaias make landfall in early October?

- (A) Near Destin, Florida
- (B) In the western Atlantic Ocean
- (C) Near the European coast
- (D) In southeastern Louisiana

2. What did traditional models first predict about Isaias?

- (A) It would stay over the ocean
- (B) It would weaken before landfall
- (C) It would hit the western Florida Panhandle
- (D) It would head farther west, toward southeastern Louisiana

3. How does an AI model work differently from a traditional model?

- (A) It asks people to guess the path
- (B) It ignores past storms completely
- (C) It uses only live satellite data
- (D) It looks for patterns by studying thousands of past storms

4. According to the article, what is a neural network?

- (A) A weather station on the coast
- (B) A set of computer programs modeled on the human brain and trained to find patterns in huge amounts of data
- (C) A kind of satellite that tracks storms
- (D) A rule for naming hurricanes

5. Who calls DeepMind the gold standard model?

- (A) Brennan
- (B) Michael Lowry
- (C) Jeff Brink
- (D) Sean Smith

6. According to Brennan, how did DeepMind do on Isaias's strength?

- (A) It refused to give a strength forecast
- (B) It was the only model used
- (C) It beat every other forecast by far
- (D) It did not do as well as the blended forecast

Answer key (try the quiz first!): 1: A 2: D 3: D 4: B 5: B 6: D

Questions to think about

1. Michael Lowry: He calls DeepMind the gold standard model and says it has been by far the best-performing computer model forecasters have ever seen.

2. Brennan: He says forecasters turn shifting outputs into clear guidance, and that DeepMind topped other models on Isaias's path but did not do as well as the blended forecast on the storm's strength.

MY THOUGHTS (at least 20 words):